1717 Wakonade Drive
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October 10, 2025

L-PI-25-039
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Prairie Island Nuclear Generating Plant, Units 1 and 2
Docket Nos. 50-282 and 50-306
Renewed Facility Operating License Nos. DPR-42 and DPR-60

Prairie Island Nuclear Generating Plant Unit 1 Licensee Event Report 2025-002-00

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits Licensee Event Report (LER) 50-282/2025-002-00 per 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v).

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

Bryan Currier
Plant Manager, Prairie Island Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Prairie Island, USNRC
Resident Inspector, Prairie Island, USNRC
State of Minnesota

ENCLOSURE

PRAIRIE ISLAND NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 50-282/2025-002-00

4 pages follow

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	

1. Facility Name		2. Docket Number	3. Page
Prairie Island Nuclear Generating Plant, Unit 1	■ 050 / ☐ 052	-282	1 OF 4

4. Title
121 Control Room Chiller Inoperable Due to Purge System Failure.

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name		Docket Number
06	21	2025	2025	- 002 -	00	10	10	2025	Prairie Island, Unit 2	■ 050	-306
									Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
Mode 1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	✓ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	✓ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		✓ 50.73(a)(2)(i)(B)	✓ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Nathan Fedora, Senior Nuclear Regulatory Engineer	612-342-8971

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	VI	CHU	T265	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
✓ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On June 21, 2025, at 17:56 CDT, Units 1 and 2 of Prairie Island Nuclear Generating Plant (PINGP) were both in Mode 1 and operating at 100 percent power when the 121 Control Room (CR) Chiller tripped on "High Refrigerant Pressure". A Past Operability Review was performed and concluded that 121 CR Chiller was considered inoperable and unable to achieve its 30-day mission time from April 22, 2025 until it tripped on June 21, 2025. In addition, during this period of inoperability of the 121 CR Chiller, there were periods when the 122 CR Chiller was also inoperable due to support system inoperability. The cause of the 121 CR Chiller tripping was a buildup of non-condensable gases due to a failed purge compressor. A temporary purge compressor skid has been installed to ensure that the non-condensable gases are removed, and that the Chiller will operate for its 30-day mission time. The 121 and 122 CR Chiller purge compressor skids are planned to be replaced with new units per an approved engineering change.

NARRATIVE

Plant Operating Conditions Prior to the Event:
Unit 1 – Mode 1, 100 percent Power.
Unit 2 – Mode 1, 100 percent Power.

EVENT DESCRIPTION

On June 21, 2025, at 17:56 CDT, Units 1 and 2 of Prairie Island Nuclear Generating Plant (PINGP) were both in Mode 1 and operating at 100 percent power. At this time, annunciator [IB] alarm 47022-0302, 121 CONTROL ROOM WATER CHILLER TRIPPED was received and outplant operators reported that the 121 Control Room (CR) Chiller (Trane Air Conditioning, PCV-1C) [VI] tripped on "High Refrigerant Pressure".

Dates And Approximate Times of Major Occurrences:

- January 17, 2025, at 10:06 CST: While performing Test Procedure 1687 - 121 CONTROL ROOM CHILLER INSPECTION, it was documented that the timer on the purge compressor (Trane Air Conditioning, PRGAA2A1BA0B) was not advancing. This was later determined to be the time when the purge compressor failed.
- April 22, 2025, at 09:31 CDT: 121 CR Chiller was considered inoperable and unable to achieve its 30-day mission time per the Past Operability Review (POR) that was completed on August 15, 2025.
- April 23, 2025, from 03:30 to 10:59 CDT: 122 CR Chiller was inoperable due to invoking Technical Specifications (TS) Limiting Conditions for Operation (LCO) 3.0.6 based on the entry of support system TS LCO 3.7.8 Condition A for 22 Diesel Driven Cooling Water Pump (DDCLP) being inoperable.
- May 03, 2025, from 15:23 to 15:28 CDT: 122 CR Chiller was inoperable due to invoking TS LCO 3.0.6. based on the entry of support system TS LCO 3.8.9 Condition A for Motor Control Center 1AB Bus 2 being inoperable.
- May 20, 2025, from 09:53 to 11:17 CDT: 122 CR Chiller was inoperable due to invoking TS LCO 3.0.6 based on the entry of support system TS LCO 3.7.8 Condition B for “B” Cooling Water Header being inoperable.
- May 22, 2025, from 22:14 to 22:30 and 22:31 to 22:43 CDT, 122 CR Chiller was inoperable due to invoking TS LCO 3.0.6. based on the entry of support system TS LCO 3.7.8 Condition A for 22 DDCLP being inoperable.
- June 21, 2025, at 17:56 CDT: The amount of non-condensable gases reached a level that permitted the 121 CR Chiller to trip on "High Refrigerant Pressure" and the Chiller was taken out of service and TS LCO 3.7.11 Condition A was entered.
- July 5, 2025, at 18:55 CDT: 121 CR Chiller was returned to service and TS LCO 3.7.11 Condition A was exited.
- August 15, 2025 [Discovery Date]: A POR was completed and concluded that 121 CR Chiller was considered inoperable from April 22, 2025 until it tripped on June 21, 2025.

TS LCO 3.7.11 pertains to the Safeguards Chilled Water System (SCWS). Under Condition A of TS LCO 3.7.11, if one SCWS loop becomes inoperable, it must be restored to operable status within 30 days. The POR determined that the 121 CR Chiller was inoperable for greater than 30 days from April 22, 2025 until it tripped on June 21, 2025. Therefore, the required action under Condition A was not met. Condition B of TS LCO 3.7.11 requires entry into Mode 3 within 6 hours if Condition A is not completed. The action to transition to Mode 3 for both Unit 1 and 2 was not completed.

Under Condition E of TS LCO 3.7.11, if both SCWS loops are inoperable while in Modes 1–4, the plant must immediately enter LCO 3.0.3. This requires the plant to be in Mode 3 within 7 hours. During the time the 121 CR Chiller was considered inoperable, the 122 CR Chiller was also inoperable for greater than 7 hours on April 23, 2025. LCO 3.0.3 was not entered, and the required transition to Mode 3 for both Unit 1 and 2 was not completed.

(EVENT DESCRIPTION continued on next page)

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Prairie Island Nuclear Generating Plant, Unit 1	■ 050 / ☐ 052	-282	2025	002	00

NARRATIVE

EVENT DESCRIPTION (continued)

Due to these missed actions, this event is reportable under 10 CFR 50.73(a)(2)(i)(B) as an operation or condition prohibited by Technical Specifications.

The combined period of approximately 9.5 hours when the 122 CR Chiller was inoperable while the 121 CR Chiller was also inoperable is reportable per 10 CFR 50.73(a)(2)(v) as an event or condition that could have prevented the fulfillment of the safety function.

EVENT ANALYSIS

SCWS has the safety related functions to maintain suitable conditions for personnel habitability in the control room, and to supply and circulate chilled water to remove thermal energy generated by safety related equipment for equipment operability. This is done by maintaining system pressure boundary and providing chilled water flow to cooling coils associated with the control room a/c units, the Unit 1 safeguards switchgear rooms, computer room, relay room and residual heat removal pits.

Normal configuration is to have one CR Chiller running and supplying chilled water to both “A” and “B” trains since SCWS is cross-tied. During a Safety Injection event, the trains separate, and both Chillers are running and supplying their respective Safeguards Chilled Water loops.

ASSESSMENT OF SAFETY CONSEQUENCES

At the time that only the 121 CR Chiller was inoperable, the 122 CR Chiller would have been able to provide the safety related functions for the SCWS.

The 121 CR Chiller successfully operated intermittently until the trip and the determination of inoperability was based on the 30-day mission time requirement. The following was the remaining calculated run time for the 121 CR Chiller during periods of inoperability of the 122 CR Chiller:

- April 23, 2025: During a 7 hour and 29-minute inoperability of the 122 CR Chiller, an estimated 29 days of run time of the 121 CR Chiller remained.
- May 3, 2025: During a 5-minute inoperability of the 122 CR Chiller, an estimated 23 days of run time of the 121 CR Chiller remained.
- May 20, 2025: During a 1 hour and 24-minute inoperability of the 122 CR Chiller, an estimated 18 days of run time of the 121 CR Chiller remained.
- May 22, 2025: During a 28-minute inoperability of the 122 CR Chiller, an estimated 16 days of run time of the 121 CR Chiller remained.

Had a design basis event occurred during these times listed above, the 121 CR Chiller would have started and operated for a significant duration. The safety consequences of a potential trip of the 121 CR Chiller 16 days post event would not be significant. A trip of the 121 Chiller would lead to an alarm in the control room and Operations would respond and utilize procedural guidance to restart the 122 CR Chiller prior to any effected areas exceeding limits. If for some reason the “A” train rooms cannot be cooled using the 122 CR Chiller, procedures are available to the Operators to provide clear actions to take to maintain room temperatures within allowable limits.

The health and safety of the public and site personnel were not impacted during this event.

NARRATIVE

CAUSE OF THE EVENT

The cause of the 121 CR Chiller tripping was a buildup of non-condensable gases due to a failed purge compressor.

Troubleshooting took place to identify possible failure of the purge compressor. Troubleshooting refuted the possibility of fouled condenser tubes, insufficient cooling water flow, and excessive air intrusion into the chiller system. No specific failure mode of the purge compressor was identified.

CORRECTIVE ACTIONS

An identical replacement for the purge compressor skid was not available. An engineering change has been approved to replace the obsolete Trane purge compressor with a skid.

The new safety related skid is not yet available. An interim, temporarily, non-safety related skid was installed on the 121 CR Chiller and on July 5, 2025, at 18:55 CDT the 121 CR Chiller was returned to service and TS LCO 3.7.11 Condition A was exited.

A compensatory measure is currently in place to run the temporarily installed purge compressor one (1) hour every two (2) weeks to maintain the operability of the 121 CR Chiller by ensuring that the non-condensable gases are removed, and the Chiller will operate for its 30-day mission time.

This compensatory measure will remain in place until the new safety related skid can be installed on the 121 CR Chiller.

The purge compressor on the 122 CR Chiller is also scheduled to be permanently modified with a new skid.

PREVIOUS SIMILAR EVENTS

There were no previous similar events in the past three years.

ADDITIONAL INFORMATION

Energy Industry Identification System (EIIS) codes are identified in the text as [XX].